IAS PRELIMS/MAINS

Ecology & Environment

A COMPREHENSIVE COVERAGE

MODULE - 36

BIODIVERSITY - 2

BIODIVERSITY

BIODIVERSITY HOTSPOTS OF INDIA

1 HIMALAYAS

- Includes the entire Himalayan region (India, Pakistan, Nepal, Bhutan and China).
- Stretching in an arc over 3,000 kilometres, this immense mountain range, covers nearly 750,000 km².

WHY HIMALAYAS ARE A HOTSPOT?

The geological, climatic and altitudinal variations in the hotspot, as well as topographic complexity, contribute to the immense biological diversity of the mountains.

	TOTAL SPECIES	ENDEMIC SPECIES
Plants	10,000	3160
Mammals	300	12
Birds	977	15
Reptiles	176	48
Amphibians	105	42
Freshwater Fishes	269	33

TYPES OF HABITATS/ECOSYSTEMS FOUND IN HIMALAYAS

- **Alluvial Grasslands** (the Terai and Dooars)
- **Subtropical Broadleaf forests** – present along the foothills of Sivaliks. (500m-1000m)
- **Eastern Himalayan Broadleaf forests** – is a temperate broadleaf forest found in the middle elevations of the Eastern Himalayas. (2000m-3000m)
- **The Western Himalayan Broadleaf forests** - is a temperate broadleaf forest extending from the Nepal, through UP, HP and J&K of India, into Pakistan. (1500m-2600m)
- **Himalayan Subtropical Pine forests** – present at lower elevations along Greater Himalayas. (1000m-2000m)
- **Subalpine Conifer forests** (3000m-4000m)
- **Alpine Meadows** (above the tree line).

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1 HIMALAYAS

TERAI AND DOOARS

- Terai is a region in southern Nepal and northern India that lies to the south of the outer foothills of the Himalayas and north of the Indo-Gangetic Plain.
- In northern India, the Terai spreads from the Yamuna River eastward across Himachal Pradesh, Haryana, Uttarakhand, Uttar Pradesh and Bihar.
- The Dooars or Duars are the alluvial floodplain grasslands in northeastern India that lie south of the outer foothills of the Himalayas and north of the Brahmaputra River basin.
- This region is about 30 km wide and stretches over about 350 km from the Teesta River in West Bengal to the Dhansiri River in Assam.
- Dooars is analogous with the Terai in northern India and southern Nepal.
- Important protected sites - Manas National Park and Kaziranga National Park in Assam, Jaldapara National Park and Buxa Tiger Reserve in West Bengal, Katarniaghat Wildlife Sanctuary in UP and Rajaji National Park in Uttarakhand.

Learning Space

ISSUES IN HIMALAYAS

1 HABITAT LOSS

- Over 75% of the Himalayas original habitat has been destroyed and degraded.
- This is mainly due to:
  a) Extensive livestock grazing has destroyed habitat compositions.
  b) Conversion of forest lands to agricultural lands.
  c) Fuel wood and fodder collection has damaged forests.

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1 HIMALAYAS

ISSUES IN HIMALAYAS

1 HABITAT LOSS

Overgrazing

- Many rural people depend on cattle for their livelihoods but do not have sufficient land for grazing. It is common to see cows, buffalo and goats grazing in forests.
- Forests can sustain a small amount of grazers, but current numbers are unsustainable. Grazers often eat saplings and destroy future forest regeneration.
- Areas in the Himalayas that have been significantly affected include the Himachal Pradesh region. HP is particularly affected by the grazing of livestock due to increasing milk production.

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ISSUES IN HIMALAYAS

2 FUEL AND FODDER COLLECTION

- Fuel and fodder collection is one of the main sources of income throughout the communities of the Himalayans.
- The need for firewood is not only a burden for nature, but also for people. Women usually spend hours in the forest looking for firewood, which increases their chances for conflict with wildlife.

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1 HIMALAYAS

ISSUES IN HIMALAYAS

3 HUMAN – WILDLIFE CONFLICT

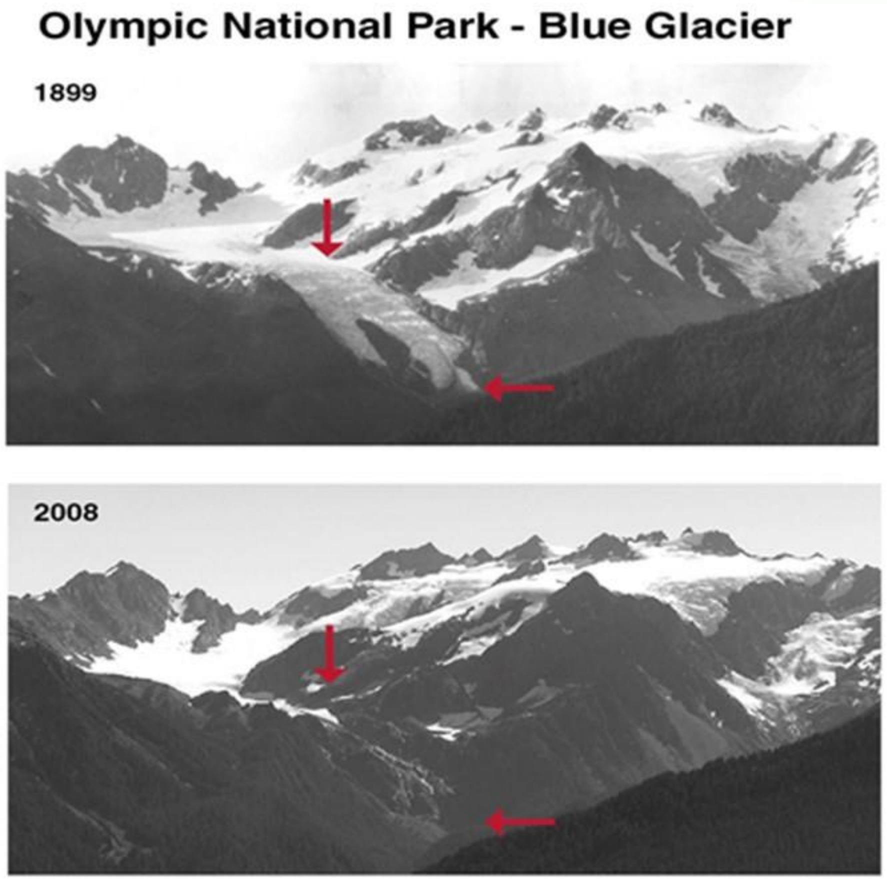
- The frequency of human-wildlife conflict increases as human populations grow and more forest land is cleared.
- Initiative of WWF
  - ✓ A community-managed livestock insurance plan compensates villagers for livestock losses from snow leopards. As a result of management of this conflict, snow leopard numbers are on the rise.

ISSUES IN HIMALAYAS

4 CLIMATE CHANGE

- Climate change is impacting people and threatening wildlife in the Himalayas. Many glaciers are melting and forming lakes which are prone to bursting and downstream flooding.
- Crops suffer from changing patterns of rainfall, which threatens the food security of the local people.
- Warmer temperatures and changing humidity have brought insect pests and disease to areas where they were previously absent.

ISSUES IN HIMALAYAS



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ISSUES IN HIMALAYAS

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